

MUHAMMAD ILHAM

DevOps / Platform Engineer - Kubernetes / Docker / Linux / Nginx - with Full-Stack Background

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PROFESSIONAL SUMMARY

DevOps-oriented engineer graduated from D4 Informatics Engineering, Politeknik Negeri Jember, with internship experience as Web Development Intern at PT. cmlabs Malang (Aug - Dec 2025, SEO-focused company). Shipped an **SEO-friendly Next.js landing page (SSR + SSG) deployed on Vercel with auto-builds** and validated delivery with **Lighthouse performance/SEO audits**. Thesis research on **ModSecurity + Nginx Caching for HTTP(S) Flood mitigation on multi-node Kubernetes** achieved a **95% block rate at 1,000 concurrent users (JMeter)** while keeping app-pod CPU **under 25 millicores**. Also builds web apps with TypeScript/Next.js and Laravel. Holds BNSP Network Administrator (2022) and Web Developer (2026) certifications.

TECHNICAL SKILLS

Infrastructure and Platforms: Kubernetes (multi-node, NGINX Ingress), Docker, Proxmox Virtual Environment (LXC/KVM), Linux server administration, AWS EC2 (thesis infra host), Vercel (auto-build deploys)

Networking and Security: Nginx reverse proxy and caching, ModSecurity WAF (OWASP CRS), TLS termination, load balancing, Apache JMeter load testing, Lighthouse performance and SEO auditing

Automation and Workflow: CI/CD concepts via Vercel auto-build deploys, Git/GitHub, containerd, kubeadm/kubectll/kubelet, Bash scripting, NFS storage

Development (supporting): TypeScript, Next.js (SSR, SSG, App Router), Laravel, Node.js (basic), REST API, PostgreSQL, SQLite, Prisma ORM

WORK EXPERIENCE

Undergraduate Thesis Researcher (DevOps / Security) - Politeknik Negeri Jember

Nov 2025 - Jun 2026 | Jember, East Java (Supervisor: Ery Setyawan Jullev Atmadji, S.Kom., M.Cs.)

- Designed a **dual-layer defense (ModSecurity WAF + Nginx Caching)** on NGINX Ingress Controller in a **multi-node Kubernetes cluster nested on Proxmox Virtual Environment (LXC/KVM)**, hosted on AWS EC2 (2 vCPU, 8 GB RAM).
- Evaluated WordPress, Ghost, and Drupal targets with **incremental load and stress tests up to 1,000 concurrent users (Apache JMeter)**; combined mitigation kept app-pod CPU **under 25 millicores** with a stable **95% attack block rate** and isolated backends from downtime.
- Documented full methodology and results in a 160-page thesis (S.Tr.Kom, 2026): single-mitigation gaps (cache bypass, Ingress CPU bottleneck) vs. combined-mitigation gains, incl. context-switch and CPU-throttling analysis on the 2-core host.

Web Development Intern - PT. cmlabs, Malang, East Java

Aug 2025 - Dec 2025 | Web Development Division (SEO-focused company)

- Built an **SEO-friendly internal landing page with Next.js (App Router, SSR + SSG)** from the internal Figma design system: 7 page groups (Homepage, About, Team, Services, Portfolio, Blog, Contact) as reusable components.
- Owned the delivery loop: **Vercel auto-build deploys on push** for live review, plus **Lighthouse audits** covering FCP/LCP, blocking time, CLS stability, responsiveness, and technical SEO (metadata, sitemap, robots).
- Transferable to DevOps/SRE work: reproducible builds, deploy-then-verify discipline, and metric-driven follow-ups (client-JS blocking time, image optimization) in a professional team workflow.

SELECTED PROJECTS

PANDAWA - Geospatial Natural Resources Information System, Bondowoso - TypeScript, Next.js, Prisma ORM, PostgreSQL

Feb 2025 - Jun 2025 | Associated with Politeknik Negeri Jember

- Built web platform to map, monitor, and manage natural-resource data for Bondowoso Regency.
- Registered software copyright No. 000942745 (Jul 2025, Directorate General of Intellectual Property).

Waves - Multi-Function Web Toolkit - *Next.js, TypeScript, FastAPI, Python*

Sep 2026 - Present | Self-initiated

- Developing a modular toolkit (REST APIs with FastAPI, TypeScript/Next.js frontend), incl. application-security and ML experiments.

Amfibi - Car Marketplace and Rental Platform - *Laravel, Livewire, Tailwind CSS, PostgreSQL*

Feb 2024 - Jun 2024 | Full-Stack Web Developer

- Built marketplace + rental (real-time chat, booking calendar, automated invoicing, RBAC); database migrated from MySQL to PostgreSQL.

RisqiShop - Cashier and Inventory Management (Desktop) - *Java, SQLite*

Feb 2023 - Present | Software Engineer (end-to-end)

- Sales/stock app with PDF/Excel reporting; database migrated from MySQL to SQLite for lightweight local deployment.

EDUCATION

D4 Informatics Engineering - Politeknik Negeri Jember - GPA 3.67

2022 - 2026 | Jember, East Java

Computer and Network Engineering - SMKN 1 Bondowoso

2019 - 2022 | Bondowoso, East Java

CERTIFICATIONS AND ACHIEVEMENTS

- **BNSP Network Administrator (KKNI Level II)** - LSP SMKN 1 Bondowoso (Jun 2022): computer network management and administration.
- **BNSP Web Developer** - LSP Politeknik Negeri Jember (Apr 2026): web application development and implementation per national competency standard.
- **Software copyright:** PANDAWA Geospatial Natural Resources Information System of Bondowoso - No. 000942745 (Jul 2025).

LANGUAGES

Indonesian - Native (professional) | English - Basic